

JIANKE YANG

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EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO, San Diego, CA Sep 2022 - Jun 2027 (Expected)

PhD in Computer Science

- Advisor: Prof. Rose Yu
- Research area: Geometric deep learning, Generative AI, AI for scientific discovery

TSINGHUA UNIVERSITY, Beijing, China

Sep 2018 - Jun 2022

Bachelor in Computer Science and Technology; Minor in Statistics

- Major GPA: 3.94/4; Rank: 5/231

PUBLICATIONS

Jianke Yang, Ohm Venkatachalam, Mohammad Kianezhad, Sharvaree Vadgama, Rose Yu. Think like a Scientist: Physics-guided LLM Agent for Equation Discovery. *Preprint*, 2026.

Jianke Yang, Manu Bhat, Bryan Hu, Yadi Cao, Nima Dehmamy, Robin Walters, Rose Yu. Discovering Symbolic Differential Equations with Symmetry Invariants. *Transactions on Machine Learning Research (TMLR)*, 2026. **(Featured Certification)**

Jianke Yang, Wang Rao, Nima Dehmamy, Robin Walters, Rose Yu. Symmetry-Informed Governing Equation Discovery. *Neural Information Processing Systems (NeurIPS)*, 2024.

Jianke Yang, Nima Dehmamy, Robin Walters, Rose Yu. Latent Space Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2024.

Jianke Yang, Robin Walters, Nima Dehmamy, Rose Yu. Generative Adversarial Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2023.

Manu Bhat, Jonghyun Park, **Jianke Yang**, Nima Dehmamy, Robin Walters, Rose Yu. AtlasD: Automatic Local Symmetry Discovery. *International Conference on Machine Learning (ICML)*, 2025.

Jianke Yang, Honglu Zhou, Kai Li, Rose Yu, Martin Renqiang Min. Physics-Informed Human Motion Generation. *In submission*, 2025.

Chuizheng Meng, **Jianke Yang**, Hao Niu, Guillaume Habault, Roberto Legaspi, Shinya Wada, Chihiro Ono, Yan Liu. Sample-Level Prototypical Federated Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025.

WORK EXPERIENCE

Research Intern, Bytedance Inc.

Jun 2024 - Sep 2024

- Latent diffusion model for protein dynamics generation

Research Intern, NEC Laboratories

Jun 2023 - Sep 2023

- Diffusion model for human motion generation
- Incorporate physics simulator into training to perform physically valid generation

RESEARCH COMMUNITY SERVICE

Program Committee

- Topology, Algebra, and Geometry in Data Science (TAG-DS) (2025)

Reviewer

- ICML (2023-2025), ICLR (2023-2025), NeurIPS (2023-2025), AISTATS (2024-2025), TMLR (2024-2025), JMLR (2024)

TEACHING

Teaching Assistant, CSE291 Generative AI, UC San Diego

Sep 2023 - Dec 2023

INVITED TALKS

- (Mar 2025) **Boston Symmetry Day**, Northeastern University
- (Nov 2024) **AAAI Fall Symposium** on Computational Approaches to Scientific Discovery
- (Oct 2023) **SciML Webinar**, University of Michigan
- (Mar 2023) **ML in Physics (Guest Lecture)**, UC San Diego

HONORS & AWARDS

- UCSD Graduate Fellowship (2022)
- Academic Excellence Award (2019, 2020, 2021) at Tsinghua University
- First Prize (2020) in China National Contest of Mathematical Modeling for Undergraduates

PROGRAMMING SKILLS

Python, R, MATLAB, C/C++, Java, JavaScript, Rust